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Pillar: Kolmogorov Theory

Wilson–Cowan and the Stuart–Landau Limit

A Bogdanov–Takens View

Why WILCO contains SL as a parameter-tuned special case

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Abstract

The Wilson–Cowan (WILCO) model is often described as a biologically grounded extension of the Stuart–Landau (SL) oscillator. We make this relationship precise. In a first-order rate-based WILCO with a sigmoidal gain, oscillation requires recurrent self-excitation: when $\kappa_{xx} = 0$ the trace of the Jacobian is strictly negative everywhere and no Hopf bifurcation is possible. With $\kappa_{xx} > 0$ above a threshold determined by the leak terms, two structurally distinct oscillator regimes emerge in the same equations, separated by a single codimension-2 Bogdanov–Takens (BT) point in parameter space. When the cross-coupling product $\kappa_{xy}\kappa_{yx}$ is small, WILCO has a clean Hopf bubble ($\mathrm{HB}^- \to \mathrm{LC} \to \mathrm{HB}^+$) whose center-manifold reduction at either end is Stuart–Landau — in this regime WILCO *is* SL. When the cross-coupling product is large, the BT point enters the oscillatory window: the SN, SNIC, and Hopf curves all spawn from it, the cusp introduces bistability, and SNIC introduces Class I excitability — in this regime WILCO is *more than* SL. Both regimes are slices through the same equations on opposite sides of the BT. We compute the bifurcation skeleton in both regimes, show that the canonical WILCO scenario (SN–SNIC–HB^+) is the BT-organized $\kappa_{xx} \approx 12$ slice at strong cross-coupling, exhibit a clean SL-equivalent slice at $\kappa_{xx} = 11$ with weak cross-coupling, and discuss the bridge to second-order neural-mass models (NMM1 / Jansen–Rit / PING), which reach the same SL skeleton via a different mechanism: synaptic phase-shift rather than recurrent gain.

Keywords: Wilson–Cowan, Stuart–Landau, Bogdanov–Takens, Hopf bifurcation, SNIC, Class I/II excitability, neural mass models, Jansen–Rit.

Classification: MSC 37G15, 37N25, 92C20

Contents

1 Introduction and one-paragraph summary

The Wilson–Cowan (WILCO) model[1] portrays cortical mass dynamics as a dialogue between excitatory and inhibitory populations interacting through synaptic weights, time-constants, sigmoid gains, and external drives. Compared with the Stuart–Landau (SL) oscillator — the universal normal form near a supercritical Hopf bifurcation — WILCO is biologically transparent, but its dynamical menu is also richer: bistability, Class I excitability, and saturation regimes that SL cannot reach. This paper makes the comparison precise.

The central observation is structural and can be stated in one sentence:

Summary

WILCO is Stuart–Landau when the cross-coupling product $\kappa_{xy}\kappa_{yx}$ is below a threshold, and SL plus a Bogdanov–Takens-organized cusp + SNIC chamber when it is above — same equations, two oscillator personalities, separated by a single codimension-2 point.

That single fact organizes everything else. Below the threshold WILCO has a clean Hopf bubble with no fold or SNIC, and the local center-manifold reduction at either Hopf is exactly the SL normal form. Above the threshold the BT point enters the oscillatory window; the saddle-node, saddle-node-on-invariant-circle, and Hopf curves all emanate from it, and a 1-parameter slice intersects them in a definite order, giving the bistability + SNIC + Hopf skeleton seen in classical Wilson–Cowan bifurcation diagrams. The same WILCO equations contain both regimes; the bifurcation parameter is the cross-coupling product, not a structural feature of the model.

A secondary but useful observation concerns the contrast with second-order neural mass models such as Jansen–Rit and PING-style networks (collectively NMM1 in our taxonomy). These models obtain a Hopf bifurcation *without* requiring strong recurrent self-excitation, because the second-order synaptic kernel supplies the destabilizing reactance internally as a phase shift around the E–I loop. The end result — a clean Hopf bubble, locally SL — is structurally equivalent to weak-cross-coupling WILCO, but the route is different: synaptic memory in NMM1 versus recurrent gain in WILCO.

The remainder of the paper unpacks these claims with explicit calculations and numerics. Section 2 fixes notation. Section 3 establishes the trace argument that no first-order rate-WILCO can oscillate without self-excitation. Section 4 maps the strong-cross-coupling regime, identifies the BT point, and reproduces the canonical SN–SNIC–HB$^+$ scenario. Section 5 maps the weak-cross-coupling regime and exhibits the clean SL bubble. Section 6 consolidates the cases as a table. Section 7 bridges to NMM1/PING. Section 8 discusses Rosetta-Stone implications.

2 The Wilson–Cowan equations

We work with the rate-based, first-order WILCO model with a single sigmoid:

$$\tau_x \dot{x} = -\gamma_x x + \sigma(\kappa_{xx}x - \kappa_{xy}y + P_x)\,, \tag{1}$$
$$\tau_y \dot{y} = -\gamma_y y + \sigma(\kappa_{yx}x - \kappa_{yy}y + P_y)\,, \tag{2}$$

where $x \in [0,1]$ is the (normalized) firing rate of the excitatory population, $y \in [0,1]$ the firing rate of the inhibitory population, and

$$\sigma(u) = \frac{1}{1+e^{-r(u-v_0)}} \tag{3}$$

is the sigmoidal input–output function with steepness r and threshold v_0. The synaptic weights κ_{xx}, $\kappa_{yy} \geq 0$ are intra-population (self-excitation, self-inhibition); κ_{xy}, $\kappa_{yx} \geq 0$ are cross-population (inhibition→excitation, excitation→inhibition). P_x, P_y are external drives. Throughout we fix

$$\tau_x = \tau_y = \gamma_x = \gamma_y = 1, \quad r = 1, \quad v_0 = 0, \quad \kappa_{yy} = 2, \quad P_y = -6, \tag{4}$$

unless otherwise stated, matching the canonical parameter set of Section 5 of the KT Rosetta Stone. The bifurcation parameter is the external drive P_x to the excitatory population. The two free structural knobs we vary are the recurrent self-excitation κ_{xx} and the cross-coupling product $\kappa_{xy}\kappa_{yx}$.

The Jacobian at any fixed point (x^*, y^*) is

$$J = \begin{pmatrix} (-\gamma_x + \kappa_{xx}\, a)/\tau_x & -\kappa_{xy}\, a/\tau_x \\ \kappa_{yx}\, b/\tau_y & (-\gamma_y - \kappa_{yy}\, b)/\tau_y \end{pmatrix}, \qquad \begin{aligned} a &\equiv \sigma'(u_x^*), \\ b &\equiv \sigma'(u_y^*), \end{aligned} \tag{5}$$

where u_x^*, u_y^* denote the values of the sigmoid arguments at the fixed point. We denote $\operatorname{tr} J$ and $\det J$ the trace and determinant. A Hopf bifurcation requires $\operatorname{tr} J = 0$ together with $\det J > 0$; a saddle-node requires $\det J = 0$.

Remark on the inhibitory drive P_y

The choice $P_y < 0$ is structurally important and is not a free convention. With $P_y \approx 0$ the inhibitory population is sufficiently driven by external input alone that y^* saturates well above zero on the lower-x branch, which pins the lower stable fixed point at $x^* \approx 0$. There $\sigma'(u_x^*) \approx 0$, so the destabilizing product $\kappa_{xx}\, \sigma'(u_x^*)$ in Eq. (7) cannot grow regardless of how large κ_{xx} becomes; the lower branch is over-damped and never undergoes a Hopf. We verified numerically that $P_y = 0$ with $\kappa_{xy} = \kappa_{yx} = 12$ produces no limit cycle for any κ_{xx} up to 20 in the displayed P_x range. The canonical $P_y \in [-6, -3]$ region keeps the lower stable FP off the saturation floor and is what makes the BT structure of Sec. 4 accessible. This is a feature of the rate-WILCO formulation: the inhibitory bias is a structural ingredient, not a cosmetic offset.

3 No oscillation without recurrent self-excitation

Setting $\kappa_{xx} = 0$ in Eq. (5) gives

$$\operatorname{tr} J\big|_{\kappa_{xx}=0} = -\frac{\gamma_x}{\tau_x} - \frac{\gamma_y + \kappa_{yy}\, b}{\tau_y} \;\le\; -\frac{\gamma_x}{\tau_x} - \frac{\gamma_y}{\tau_y} \;<\; 0 \tag{6}$$

uniformly in P_x, P_y, and the operating point. The trace is bounded strictly below zero by the leak terms alone. Consequently:

Proposition 1 (No-Hopf-without-recurrence)

In the first-order rate WILCO of Eqs. (1)–(2) with $\gamma_x, \gamma_y, \tau_x, \tau_y, \kappa_{yy} > 0$ and $\kappa_{xx} = 0$, no fixed point can undergo a Hopf bifurcation. The system has a unique stable fixed point for every (P_x, P_y), given by the simultaneous solution of $x = \sigma(-\kappa_{xy} y + P_x)$ and $y = \sigma(\kappa_{yx} x - \kappa_{yy} y + P_y)$, both monotone in their respective variables.

This has two structurally important consequences. First, it shows that the intuition that "bias alone suffices" — positioning the operating point on the steep part of σ via P_x — is necessary but *not* sufficient. The bias enters the trace only through the product $\kappa_{xx}\, \sigma'(u_x^*)$, and when

$\kappa_{xx} = 0$ the bias has no leverage on $\operatorname{tr} J$ whatsoever. The correct statement of Ermentrout's criterion[2] is therefore an "and", not an "or":

$$\boxed{\frac{\kappa_{xx}\,\sigma'(u_x^*)}{\tau_x} > \frac{\gamma_x}{\tau_x} + \frac{\gamma_y + \kappa_{yy}\,\sigma'(u_y^*)}{\tau_y}} \tag{7}$$

Both ingredients matter: $\kappa_{xx} > 0$ to provide the destabilizing direction at all, and the bias P_x to position the operating point so $\sigma'(u_x^*)$ is large.

Second, this is exactly the structural difference between rate-based WILCO and second-order neural mass models such as Jansen–Rit and PING: in those models, the second-order synaptic kernel supplies a destabilizing phase shift internally and the trace inequality (7) is replaced by a *loop-gain* condition that does not require $\kappa_{xx} > 0$. We return to this in Sec. 7.

Figure 1 illustrates the proposition numerically. Panel (a) shows the $\kappa_{xx} = 0$ slice: a single stable fixed point, monotone sigmoidal in P_x, with no oscillation anywhere in the displayed range. Panel (b) shows that even at $\kappa_{xx} = 6$ the system is still below the Hopf threshold for the present parameters and remains monostable. Panel (c) is the canonical setting $\kappa_{xx} = 12$, where the full SN–SNIC–HB$^+$ sequence appears.

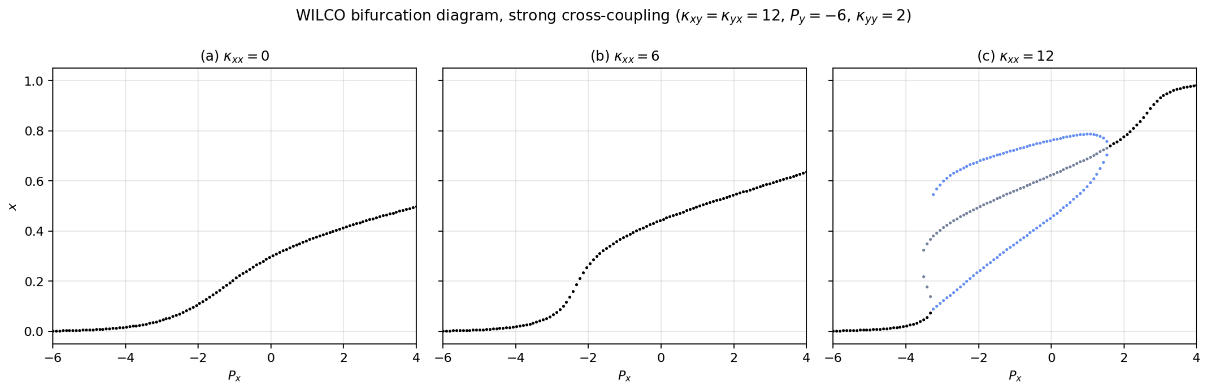

Figure 1: Bifurcation diagram of WILCO under variation of P_x for three values of the recurrent self-excitation κ_{xx}. Full parameter set: $\tau_x = \tau_y = \gamma_x = \gamma_y = 1$, sigmoid steepness $r = 1$, threshold $v_0 = 0$, $\kappa_{xy} = \kappa_{yx} = 12$ (strong cross-coupling), $\kappa_{yy} = 2$, $P_y = -6$. (a) $\kappa_{xx} = 0$: monotone stable fixed point everywhere; no oscillation possible (Proposition 1). (b) $\kappa_{xx} = 6$: still below the Hopf threshold; FP curve sigmoidal but stable. (c) $\kappa_{xx} = 12$: full SN–SNIC–HB$^+$ scenario. Black: stable fixed point. Gray: unstable fixed point. Blue: limit-cycle minima/maxima.

4 Strong cross-coupling: the Bogdanov–Takens regime

We now fix $\kappa_{xy} = \kappa_{yx} = 12$ (cross-coupling product 144) and explore the (κ_{xx}, P_x) plane. Figure 2(a) shows the result of a two-parameter scan in which we count fixed points and classify their stability at each grid cell, then test for a limit cycle whenever no fixed point is stable.

The colored region in Fig. 2(a) is the union of all parameter values supporting some structure beyond a single stable fixed point. It separates into:

- **Limit-cycle region (light blue, regime 2):** a single unstable fixed point plus a stable limit cycle. This is the territory in which the system is "purely oscillating".
- **One-FP-plus-LC region (light green, regime 4):** three fixed points (a stable node at one end, a saddle, and an unstable focus surrounded by a limit cycle). This is the geometry

of the bistable + cycle coexistence one crosses through on the way out of the cycle in the canonical Fig. 5.1 ordering.

- **Bistable-FP region (light pink, regime 3):** three fixed points, two stable, separated by a saddle. Cusp territory with no oscillation.

The colored regions converge toward a single point at the lower edge of the colored island: this is the **Bogdanov–Takens (BT) point**, the codimension-2 organizing center from which the SN curve, the Hopf curve, and the SNIC/homoclinic curve all emanate. As κ_{xx} is increased from below, a BT point enters the oscillatory window and three bifurcation curves spawn from it. Crucially, a 1-parameter slice in P_x at fixed κ_{xx} above the BT *must* cross all three loci, in an order that depends on which side of the BT one is.

The canonical slice ($\kappa_{xx} = 12$, above BT). Panel (c) of Fig. 2 shows the slice. Reading P_x left-to-right one encounters

white (1 stable FP) $\rightarrow$ green (3 FPs + LC) $\rightarrow$ blue (1 unstable FP + LC) $\rightarrow$ white,

which translates into the bifurcation sequence

$$\text{SN} \;\rightarrow\; \text{SNIC} \;\rightarrow\; \text{HB}^{+}. \tag{8}$$

This is precisely the topology of the canonical Wilson–Cowan bifurcation diagram: a saddle-node creates a saddle plus an unstable focus surrounded by a limit cycle (entry to green); the lower stable node and the saddle subsequently collide *on* the limit cycle (SNIC, exit of green into blue); the limit cycle finally shrinks onto the upper unstable focus and dies via a supercritical Hopf (HB$^+$, exit of blue into white). The center-manifold reduction at HB$^+$ is the Stuart–Landau normal form. The local reduction near SNIC is the theta-neuron / quadratic integrate-and-fire normal form[2], characterized by Class I excitability (frequency $\propto \sqrt{\mu - \mu_c} \to 0$ at onset). The local reduction near SN is the saddle-node normal form (no oscillation by itself).

The below-BT slice ($\kappa_{xx} = 8$, below BT). Panel (b) of Fig. 2 shows the slice immediately below the BT tip. With these parameters the slice does not reach the colored region at all — it remains in white throughout the displayed P_x range. This is consistent with Eq. (7): at $\kappa_{xx} = 8$ the destabilizing product $\kappa_{xx}\sigma'$ does not exceed the leak terms even at the optimally-positioned operating point.

Local normal forms. Each of the three bifurcations has its own canonical reduction, and the SL approximation is *not* uniformly valid across the cycle:

- Near **HB$^+$**: Stuart–Landau, $\dot{z} = (\mu + i\omega)z - |z|^2 z$. Class II excitability.
- Near **SNIC**: theta-neuron / QIF, $\dot{\theta} = (1 - \cos\theta) + (1 + \cos\theta)\,\mu$. Class I excitability.
- Near **SN**: saddle-node, $\dot{x} = \mu + x^2$. Bistability/cusp; no oscillation by itself.

All three are local approximations to *different corners* of the BT unfolding[2]. The "WILCO reduces to SL near criticality" statement common in the literature is correct, but "criticality" must be interpreted as Hopf criticality at HB$^+$, not as the entire transition region.

5 Weak cross-coupling: the pure Hopf bubble

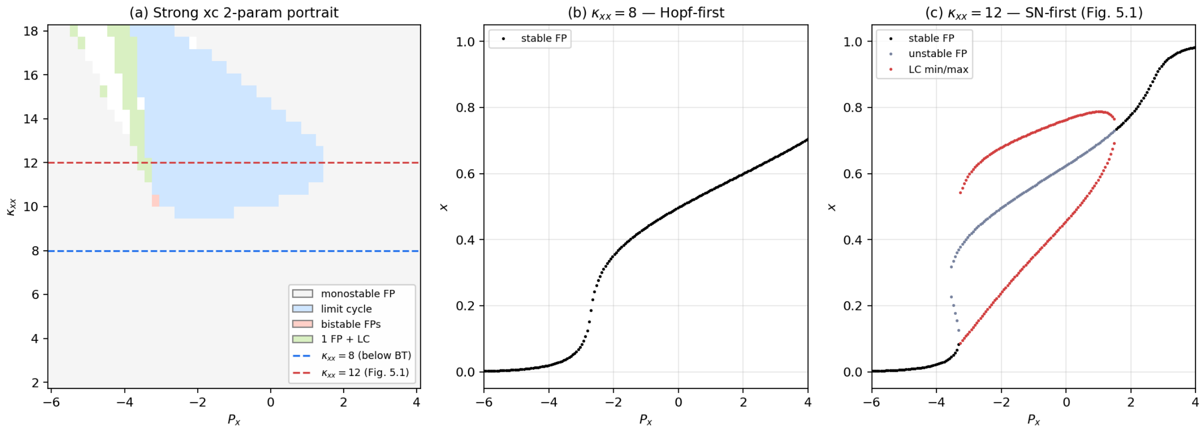

Figure 2: Two-parameter portrait of WILCO in the (κ_{xx}, P_x) plane at *strong* cross-coupling, with two slices. Full parameter set: $\tau_x = \tau_y = \gamma_x = \gamma_y = 1$, sigmoid steepness $r = 1$, threshold $v_0 = 0$, $\kappa_{xy} = \kappa_{yx} = 12$, $\kappa_{yy} = 2$, $P_y = -6$. (a) Regime classification: white = monostable FP; light blue = limit cycle (regime 2); light green = 1 stable FP + LC + saddle + unstable focus (regime 4); light pink = bistable FPs (regime 3). The Bogdanov–Takens (BT) point sits at the lower tip of the colored region. (b) Slice at $\kappa_{xx} = 8$, below the BT tip: monostable everywhere. (c) Slice at $\kappa_{xx} = 12$, above the BT, reproducing the canonical Wilson–Cowan SN $\to$ SNIC $\to$ HB$^+$ ordering as P_x increases.

We now repeat the two-parameter scan with a weaker cross-coupling, $\kappa_{xy} = \kappa_{yx} = 8$ (cross-coupling product 64). Figure 3(a) shows the result. Two structural changes compared with Fig. 2(a):

1. The bistable wedge (light pink) and the limit-cycle region (light blue) have *detached* — they no longer share a boundary.
2. The "one-FP-plus-LC" mixed region (light green) is essentially absent in the displayed window.

Geometrically: the SN/cusp wedge has shrunk and migrated to higher κ_{xx}, while the Hopf bubble persists at intermediate κ_{xx}. The BT point that organized the strong-coupling regime has effectively "left" the oscillatory window. This is what we mean by saying that the cusp and the Hopf bubble are independent ingredients, controlled by different parameter products.

The slice at $\kappa_{xx} = 11$, panel (b) of Fig. 3, traces a clean Hopf bubble:

$$\text{white (stable FP)} \;\to\; \text{HB}^- \;\to\; \text{blue (LC)} \;\to\; \text{HB}^+ \;\to\; \text{white (stable FP)}.$$

The cycle is born of small amplitude at HB$^-$, grows smoothly across the bubble, and dies of small amplitude at HB$^+$. There is no fold, no SNIC, no bistability anywhere on the slice. The center-manifold reduction at *both* ends of the bubble is the Stuart–Landau normal form. This is the WILCO realization of an SL oscillator. Class II excitability only. Note that for this figure we used $P_y = -3$ to place the inhibitory operating point in a regime where the cusp wedge is well separated from the Hopf locus; the qualitative picture — pure bubble achievable by weakening cross-coupling — is robust to the choice of P_y.

Three personalities, side by side. Figure 4 contrasts the three structurally distinct attractors of WILCO — monostable rest, BT-regime cycle, and pure Hopf bubble — as phase-plane trajectories (top row) and $x(t)$ time series (bottom row). The visual contrast is meaningful: only

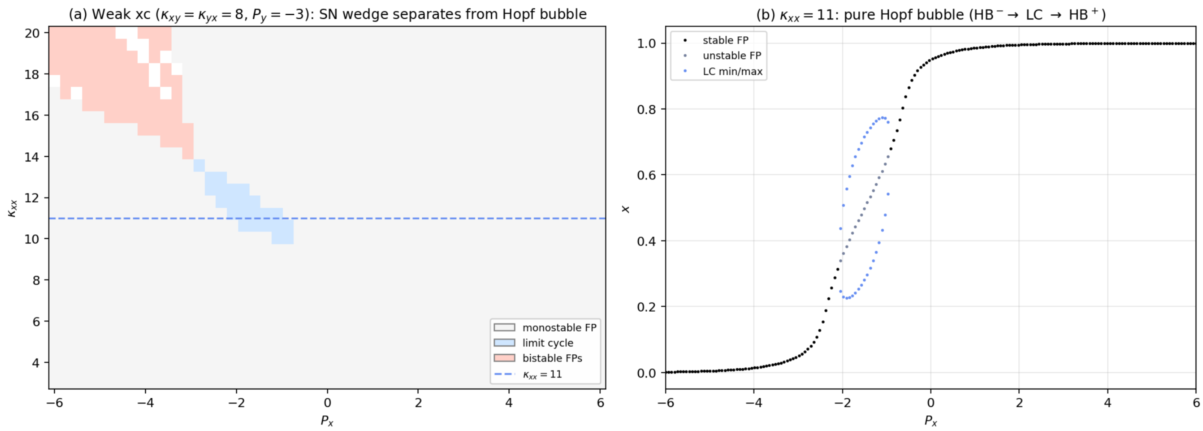

Figure 3: Two-parameter portrait of WILCO at *weak* cross-coupling, with a slice through the pure Hopf bubble. Full parameter set: $\tau_x = \tau_y = \gamma_x = \gamma_y = 1$, sigmoid steepness $r = 1$, threshold $v_0 = 0$, $\kappa_{xy} = \kappa_{yx} = 8$, $\kappa_{yy} = 2$, $P_y = -3$ (note: deviates from the canonical $P_y = -6$; see the remark in Sec. 2). (a) The bistable wedge (light pink) and the limit-cycle region (light blue) are detached — the cusp/SN locus and the Hopf locus no longer share a boundary. (b) Slice at $\kappa_{xx} = 11$: a clean pure Hopf bubble bounded by HB^- and HB^+, no SN or SNIC. The center-manifold reduction at both ends of the bubble is Stuart–Landau.

the no-oscillation case (a) and the SL-bubble case (c) collapse onto the same SL-style template; the BT-near-SNIC case (b) is qualitatively different. The cycle in (b) has a long quiescent phase near the location where the saddle and node have just collided, producing a pulse-like time series with sharply defined firing events — the canonical Class I excitability signature, formally captured by the theta-neuron normal form. The cycle in (c) is near-sinusoidal, uniform in amplitude and period — the canonical Class II / Stuart–Landau signature. Both cycles are limit cycles in the same E–I phase space, but their *waveforms* encode the different normal forms governing them.

6 Summary of cases

Table 1 consolidates the cases discussed and adds the NMM1/PING comparison developed in Sec. 7.

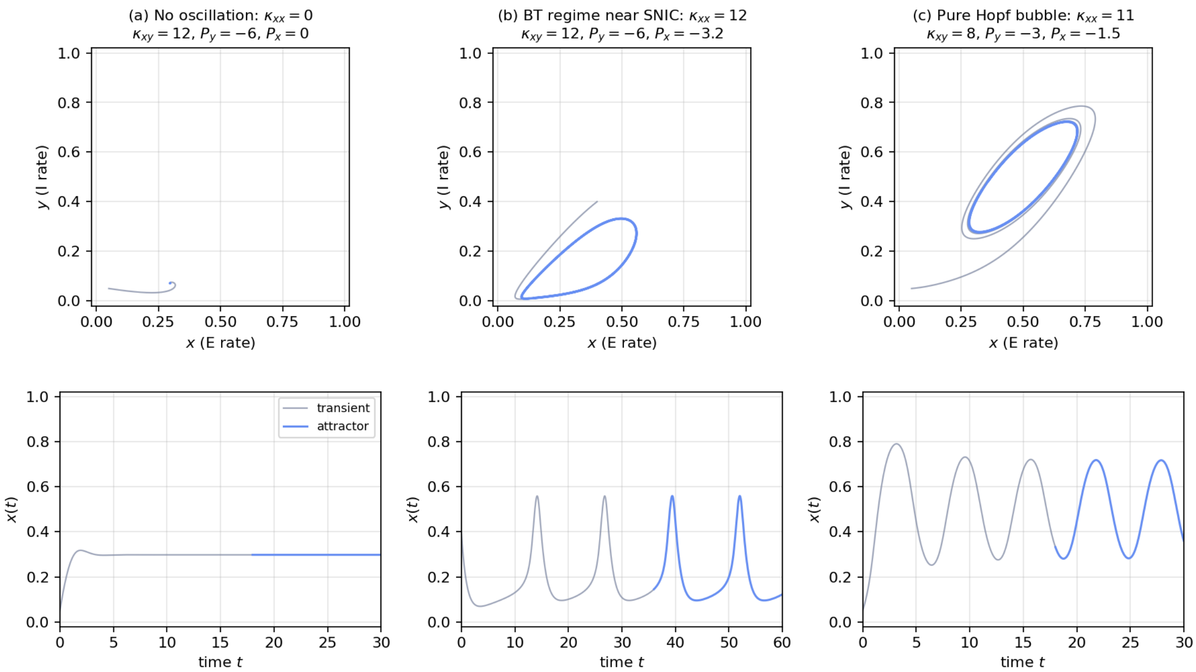

Figure 4: Phase-plane trajectories (top row) and $x(t)$ time series (bottom row) for the three WILCO regimes. Shared parameters across all panels: $\tau_x = \tau_y = \gamma_x = \gamma_y = 1$, sigmoid steepness $r = 1$, threshold $v_0 = 0$, $\kappa_{yy} = 2$. The remaining parameters change between panels. (a) No oscillation: $\kappa_{xx} = 0$, $\kappa_{xy} = \kappa_{yx} = 12$, $P_y = -6$, $P_x = 0$; the orbit spirals into a stable fixed point. (b) BT regime near SNIC: $\kappa_{xx} = 12$, $\kappa_{xy} = \kappa_{yx} = 12$, $P_y = -6$, $P_x = -3.2$ (just past SNIC); a long-period limit cycle with a slow phase near the saddle-node ghost — Class I excitability, pulse-like waveform. (c) Pure Hopf bubble: $\kappa_{xx} = 11$, $\kappa_{xy} = \kappa_{yx} = 8$, $P_y = -3$, $P_x = -1.5$ (mid-bubble); a smooth, uniformly-shaped limit cycle — Class II excitability, near-sinusoidal waveform, locally Stuart–Landau. Gray segments are transients; blue segments show the asymptotic attractor.

Table 1: WILCO regimes and their oscillator personalities, plus the NMM1/PING comparison. “xc” denotes the cross-coupling product $\kappa_{xy}\kappa_{yx}$.

#	κ_{xx}	xc	Order along P_x	Bifurcations	Local reductions / Excitability
1	0	any	white only	none	Over-damped; no oscillation.
2	sub-Hopf	strong	white only	none	Below threshold of Eq. (7).
3	≈ 11	64 (weak)	white → blue → white	HB^-, HB^+	SL at both ends. Class II.
4	≈ 8–10	144 (strong)	white → blue → green → white	HB^-, SN, SNIC	SL → fold → θ-neuron.
5	≈ 12	144 (strong)	white → green → blue → white	SN, SNIC, HB^+	fold → θ-neuron → SL. Class I and II.
6	not needed	—	white → blue → white	HB^-, HB^+	SL at both ends. Class II. NMM1/PING/JR.

Cases 4 and 5 are the two halves of the Bogdanov–Takens unfolding at strong cross-coupling: the same three bifurcations in opposite order along P_x, with the BT point sitting between them. As κ_{xx} is increased through the BT, the order in which a P_x-sweep encounters SN, SNIC, and Hopf flips. Cases 3 and 6 are equivalent at the level of the bifurcation skeleton: both are pure Hopf bubbles with SL reductions at both ends. They differ only in *how* the destabilizing reactance is supplied — recurrent gain in case 3, second-order synaptic kernel in case 6.

7 Bridge to NMM1: a different route to the same SL skeleton

The Jansen–Rit-style neural mass models (NMM1 in our taxonomy) replace the first-order synapse of WILCO with a second-order alpha-function kernel:

$$\ddot{u} + 2\alpha\,\dot{u} + \alpha^2\,u = \alpha\,A\,\sigma(\text{presynaptic input}), \tag{9}$$

which acts as a linear bandpass on the firing-rate input and supplies a smooth phase shift around the E–I loop. As a consequence, a 4D (E,I) NMM1 system can support a Hopf bifurcation at vanishing intra-population self-excitation: the destabilizing reactance comes from the synaptic kernel itself, not from κ_{xx}. The PING gamma rhythm is the canonical example.

This means that the NMM1/PING/JR Hopf bubble and the weak-cross-coupling WILCO Hopf bubble (Sec. 5) are *structurally equivalent* but *mechanistically distinct*: both reduce locally to SL near onset, both are Class II resonators with a clean pure Hopf bubble, both lack SN/SNIC structure in their canonical operating regime — but one purchases the destabilizing reactance through recurrent gain $\kappa_{xx}\sigma'$, the other through synaptic phase memory.

This dichotomy is structurally informative for whole-brain modeling. WILCO units in the strong-cross-coupling regime can produce both Class I (slow, integrator-like, near SNIC) and Class II (fast, resonator-like, near HB^+) excitability in the same 1-parameter family. NMM1 units in their canonical regime produce only Class II. Hybrid models that combine intra-region NMM1 dynamics with inter-region WILCO-style coupling inherit this taxonomy: the local rhythms are NMM1-Hopf, the network-level dynamics under variation of effective coupling can pass through BT-organized regimes.

8 Discussion: implications for the Rosetta Stone

Three statements survive the analysis as structural anchors for a Rosetta-Stone treatment of WILCO.

(i) WILCO contains SL as a parameter-tuned special case. The center-manifold reduction at a Hopf bifurcation is universal: in the weak-cross-coupling regime, WILCO and SL have the same bifurcation skeleton on the cycle. The SL reduction is local in two senses — local on the center manifold near each Hopf, and local in parameter space within the pure-bubble window.

(ii) WILCO becomes "more than SL" only above a coupling threshold. The cusp + SNIC chamber of the strong-cross-coupling regime is a genuinely new structure beyond SL: it brings bistability (cusp), Class I excitability (SNIC), and a BT organizing center that ties three bifurcations into a single unfolding. None of this is present in SL. None of it is present in weak-coupling WILCO either. The threshold is in the cross-coupling product $\kappa_{xy}\kappa_{yx}$.

(iii) WILCO and NMM1 reach the same SL skeleton via different mechanisms. The recurrent-gain route (WILCO weak-cross-coupling) and the synaptic-phase route (NMM1/PING/JR) are mechanistically distinct but structurally equivalent at the level of the Hopf normal form. This justifies treating both as "Stuart–Landau realizations" while keeping the underlying biology disambiguated. Both then sit at the same node in the Rosetta Stone graph: a clean Class II resonator with no fold structure.

The single cleanest sentence for the Rosetta Stone entry is therefore:

Section opener for §5 of the Rosetta Stone

WILCO contains the Stuart–Landau oscillator as one of its parameter-tuned limits. When the cross-coupling is weak, the model supports a clean Hopf bubble ($\mathrm{HB}^- \to \mathrm{LC} \to \mathrm{HB}^+$) with no fold or SNIC anywhere on the slice, and the center-manifold reduction at either end is the Stuart–Landau normal form. *In this limit WILCO is SL.* Once the cross-coupling product $\kappa_{xy}\kappa_{yx}$ crosses a threshold set by the diagonal terms, a Bogdanov–Takens point enters the oscillatory window. The saddle-node, SNIC, and Hopf curves emerge from it as three branches of a single codimension-2 unfolding: the cusp brings bistability, the SNIC brings Class I excitability, and the system displays dynamics genuinely beyond SL's reach. *In this regime WILCO is more than SL.* The two regimes are 1-parameter slices through the same equations, separated by exactly one codimension-2 point in parameter space. This is what gives WILCO its place in the Rosetta Stone: it specializes to SL — and to the PING/Jansen–Rit gamma rhythm — at one limit, extends to bistability and Class I excitability at the other, and the hinge between them is a single locatable point.

9 Reproducibility

All figures in this paper are produced by a single Python script, `figures/make_figures.py`, that performs the regime classification on a (κ_{xx}, P_x) grid, locates fixed points by Newton iteration from a multi-seed initial grid, classifies their stability via the eigenvalues of Eq. (5), and integrates the ODE system to recover limit-cycle minima/maxima where no fixed point is stable. Resolution knobs (`GRID_KAPPA`, `GRID_PX`, `P_FINE_N`, `SEEDS_2D`) are exposed at the top of the script for higher-quality runs locally. The script depends only on `numpy`, `scipy`, and `matplotlib`.

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